r/NoStupidQuestions

How exactly do LLMs like ChatGPT "consume" fresh water?

Posted by u/jerodimus • • 0 points • 24 comments

If the water is mainly used for cooling, can't it be reused afterwards? Is this coolant water really equivalent to fresh potable water? If so, couldn't another water source like e.g. greywater be used?

Comments

u/MonkeyBrains09 • 2 points • 2025-04-14 23:50 UTC

It eventually will make it back. Some places used closed loop cooling where the water never is lost except through leaks.

It's not like water is destroyed and even then it turns back into water eventually through the water cycle.

u/Avery_Thorn • 1 points • 2025-04-14 23:43 UTC

This is one of those things that bugs the !@*# out of me.

River water is perfectly fine for evaporative cooling. It works out nicely. There are a bunch of cloud factories where they create clouds and electricity all up and down the major rivers here. Yes, it needs a bit of filtering. No, if you use a lot of additives, you can't dump that back into the river. But you don't have to, you have to keep adding to the pool because most of the water leaves the factory, quite cleanly, up the cooling tower.

It is wasteful and stupid to use ground water for that purpose. And I cannot fathom why in the world they allow companies to build data centers in places where they would use well water for that system, when there are absolutely plenty of places where there is plenty of river water for this use.

Look, the problem is that the west has exceptionally bad water management systems, and they have fucked over their water ecosystems by doing too much with too little. In my opinion, the crops on a land should have lower water requirements than the water that the land gets. If you don't get enough water to raise cows, then perhaps that's not a great place to raise cows. If you don't get enough water for your almonds, perhaps that's not a good place to grow almonds. If you don't get enough water for your rice... perhaps growing rice in the desert is fucking idiotic.

And perhaps building a data center that needs evaporative water cooling in the middle of a desert is a bad idea, too.

Not like you can't easily ship the data from Ohio to the internet just as well as you can from Nevada.

u/MonkeyBrains09 • 1 points • 2025-04-14 21:00 UTC

It can be re-used depending on the system.

Generally the cool water is used to absorb heat from the electronics and the warm water is discharged or cooled down and reused. Cooling the water back down can be expensive so its sometimes easier and cheaper to use new water that is already cool.

u/jerodimus • 1 points • 2025-04-14 23:38 UTC

So the water isn't actually lost?

u/MonkeyBrains09 • 1 points • 2025-04-14 23:45 UTC

Water is never lost. It just goes back into the water cycle.

For a real word example. At work, we get cold water pipes in from some company to to cool out data center. The warm water is pumped back out to the same company. They use that warmer water for steam boilers to produce energy. It's more efficient for them to make steam from warm water than from cold water. It's a constant flow that never stops unless work needs to be done on the pipes.

u/jerodimus • 1 points • 2025-04-14 23:46 UTC

OK so the water used to cool AI data centres isn't actually lost or consumed, it goes back to the water cycle. So what's the big deal?

u/archpawn • 2 points • 2025-04-14 20:12 UTC

Evaporation cooling, both in terms of cooling the datacenters and the power plants.

That said, it sounds like a lot of water but it's nothing in comparison to how much is consumed in producing beef. Turns out, it takes a lot of water to grow the food you feed your food.

u/Inevitable-Regret411 • 1 points • 2025-04-14 19:33 UTC

The way cooling works is cold water is pumped through the servers, and as it flows through the pipes the heating generated by the machines flows into the water. This is because heat normally flows from where it's hotter to where it's cooler. This warms up the water to the same temperature as the machines, so it can't then be used for cooling again straight away until it cools back down. Hence the constant need for more cool water.

u/jerodimus • 1 points • 2025-04-14 23:35 UTC

But then the warm water will cool and can be used again, no?

u/Inevitable-Regret411 • 1 points • 2025-04-15 20:53 UTC

It will, but that's not cost effective. The servers that the AI runs on are active 24/7, they need a constant supply of new cool water. If you pump all the hot water into a tank and wait for it to cool down, you still need more water while you're waiting for it to cool.

u/jerodimus • 1 points • 2025-04-15 20:56 UTC

OK, but ultimately is any of the water lost?

u/jasa55 • 1 points • 2025-06-08 20:16 UTC

It's lost to the rest of the world, right.. even though the exact molecules of water may be recycled, these systems are keeping and using some X litres of water, and X is a huge amount. Just like we humans have some 30-50 litres of water in our body at all times.

u/Alesus2-0 • 3 points • 2025-04-14 19:29 UTC

My understanding is that a ChatGPT primarily runs on Microsoft servers. As I understand it, their major server farms use direct evaporative cooling systems. The air going into the server rooms is pumped through a waterlogged media. The water absorbs heat from the air and evaporates, raising humidity but dropping the air temperature. The system isn't closed, so all the water used in cooling just evaporates away. It consumes a lot of water, but is cheaper to power and maintain on a large scale than giant AC units.

u/JustSomeGuy_56 • 1 points • 2025-04-14 19:26 UTC

In the late 1970s the insurance company I worked for built a new building and (partially) heated it with the cooling water from our big IBM mainframes. We got some sort of award from the EPA.

u/FriendlyCraig • 1 points • 2025-04-14 19:18 UTC

Water in coming setups can often be treated to improve safety and thermal performance. Even if not treated warm water running over copper or aluminum will leech a bit of metal. Greywater would be a horrible idea as it would damage the internals of a cooling setup, such as by clogging or eating away at the material.

u/LuinAelin • 12 points • 2025-04-14 19:18 UTC

The water does technically return into the air and stuff.

But it still could be better used than generating an image of yourself as a doll because water used in AI isn't dunk or used on crops

> **u/jerodimus** • 1 points • 2025-04-14 23:36 UTC
>
> OK, but if it returns to the air etc. it will fall as rain or form as condensation somewhere else, right? The water isn't actually lost?

u/HeartOn_SoulAceUp • 4 points • 2025-04-14 19:16 UTC

Because the water is often pumped from clean, fresh underground aquifers...

...and released above ground... becoming contaminated groundwater... making it's way into streams and rivers.

> **u/jerodimus** • 1 points • 2025-04-14 23:37 UTC
>
> Contaminated with what?
>
> > **u/HeartOn_SoulAceUp** • 1 points • 2025-04-14 23:43 UTC
> >
> > Pesticides, car oil, antifreeze, chemical fertilizers, weed killer, etc.,
> >
> > Anything in the environment that moving groundwater picks up and carries to the ocean.
> >
> > > **u/jerodimus** • 1 points • 2025-04-14 23:45 UTC
> > >
> > > Does the data center contaminate the water? It just flows through tubes right?
> > >
> > > > **u/HeartOn_SoulAceUp** • 1 points • 2025-04-14 23:58 UTC
> > > >
> > > > Right, but unlike rainfall, which is dispersed onto every square inch of soil (allowing it to percolate into the aquifers), the huge flow of their water release never gives it the chance to do so.
> > > >
> > > > To be honest, modern cities (concrete jungles) do the same thing by not letting rainfall naturally absorb into the ground, but channel it into concrete runoff waterways.

So it's a losing battle, all around. Some cities are taking active measures to avoid this by creating places where the runoff can be slowed down and absorbed by conducive soil types, plants, rocks, and other substrates.

You asked a specific question, and I gave you a specific answer as to why LLM's consume "fresh" water.

:)

u/jerodimus • 1 points • 2025-04-15 00:00 UTC

Yes, and I appreciated it. I'm still not sure how it's actually "consumed" though, since no water is lost...

u/HeartOn_SoulAceUp • 2 points • 2025-04-15 00:29 UTC

Limited fresh water from shrinking aquifers is lost that could be used for households and farmers. It is converted into useless and contaminated runoff groundwater.

I guess that would be the argument.

:)

u/LeatherChaise • 2 points • 2025-04-14 19:16 UTC

I'd be happy if they connected it to the Dallas sanitation sewage system. Every time a Texan flushes, Elon Musk or Mark Zuckerburg can cool down.